Lecture 4
Complex Numbers

A complex number is the form $a + ib$ where a and b are real numbers and i is called the imaginary unit, has the property that $i^2 = -1$ or $i = \sqrt{-1}$. In general, if c is any positive number, we would write:

$$\sqrt{-c} = i\sqrt{c}$$

If $z = a + ib$, then a is called the real part of z and b is called the imaginary part of z and are denoted by $\boldsymbol{Re\{z\}} = a$ and $\boldsymbol{Im\{z\}} = b$ respectively. From this, it is obvious that two complex numbers $a + ib$ and $c + id$ are equal if and only if $a = c$ and $b = d$, that is, the real and imaginary components are equal. If $a = 0$ the number $z = ib$ is said to be purely imaginary, if $b = 0$ the number $z = a$ is real.
The standard rectangular form of a complex number is $z = a + ib$. The symbol z, which can stand for any complex numbers, is called a complex variable.

Graphical Representation of Complex Number/ Argand Diagram:
Since a complex number $a + ib$ can be considered as an ordered pair of real numbers, we can represent such numbers by points in a xy plane called the **complex plane or Argand diagram**. Mathematician Argand represented a complex number in a diagram known as **Argand diagram**.

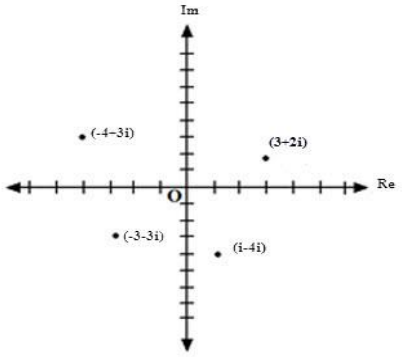


Figure1: Complex number on the Argand diagram

A complex number $a + ib$ can be represented by point P whose co-ordinates are (a, b).The horizontal axis is called the **real axis,** and the vertical axis is called the **imaginary** axis.
To each complex number there corresponds one and only one point in the plane, and conversely to each point in the plane there corresponds one and only one complex number. Because of this we often refer to the complex number z as the point. This is shown in **Figure1.**

Fundamental operations with complex numbers:
Addition and Subtraction:
The **sum** and **difference** of complex numbers is defined by adding or subtracting their real components where $a, b \in \mathbb{R}$ i.e.:

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$
$$(a - bi) + (c - di) = (a - c) + (b - d)i$$

For instance: $(3 + i) + (1 - 7i) = (3 + 1) + (1 - 7)\ i = (4 - 6i)$ (like vector addition)
Product:
The commutative and distributive properties hold for the **product** of complex numbers i.e.,

$$(a + bi)(c + di) = a(c + di) + bi(c + di) = ac + adi + bci + bdi^2$$

We know: $i^2 = -1$

Therefore, giving us: $(a+bi)(c+di) = (ac-bd) + i\ (ad+bc)$.

Division:

$$\frac{(a+bi)}{(c+di)} = \frac{(a+bi)}{(c+di)}.\frac{(c-di)}{(c-di)} = \frac{(ac+bd)}{(c^2+d^2)} + i\ \frac{(bc-ad)}{(c^2+d^2)}.$$

Basically, when **dividing** two complex numbers we are rationalizing the denominator of a rational expression multiplying the numerator and denominator by the conjugation of the denominator.

Example 4.1

Express $\frac{-3+i}{7-3i}$ in the form $a+ib$.

Solution: We must multiply the numerator and denominator by the conjugate of $7-3i$ i.e., $7+3i$.

$$\frac{-3+i}{7-3i} = \frac{-3+i}{7-3i}.\frac{7+3i}{7+3i} = \frac{(-21-3)}{(49+9)} + i\ \frac{(7-9)}{(49+9)} = -\frac{12}{29} - i\ \frac{1}{29}.$$

Conjugates

The complex conjugate, or briefly conjugate, of a complex number $z = a+ib$ is $\overline{z}$ $\overleftrightarrow{}$ $= a-ib$. The complex conjugate of a complex number z is often indicated by $\overline{z}$ $\overleftrightarrow{}$. The geometric interpretation of a complex conjugate is the reflection along the real axis. This can be seen in the **Figure 2** below where $z = a+ib$ is a complex number. Listed below are also several properties of conjugates.

Some Properties:

1) $\bar{\bar{z}} = z$
2) $\overline{z+w} = \bar{z} + \bar{w}$
3) $\overline{zw} = \bar{z}\bar{w}$
4) $\overline{\left(\frac{z}{w}\right)} = \frac{\bar{z}}{\bar{w}}, w \neq 0.$

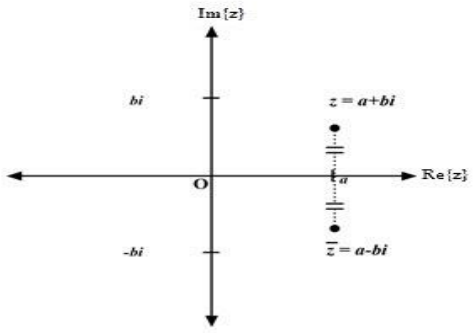


Figure 2: Complex Conjugate

Absolute value/Modulus

The distance from the origin to any complex number is the **absolute value** or **modulus.** Looking at Figure 3 below we can see that Pythagoras' Theorem gives us a formula to calculate the absolute value of a complex number $z = a+bi$ denoted by mod z or $|z|$.

i.e. mod $z = |z| = \sqrt{a^2+b^2}$

Example 4.2

If $z_1 = 1-i$, $z_2 = -2-3i$ and $z_3 = 2i$, then evaluate $|3{z_1}^3 + 2\overline{z_2} - 5z_3|$.

Solution:

Given, $z_1 = 1-i$, $z_2 = -2-3i$, $z_3 = 2i$

$$\therefore |3{z_1}^3 + 2\overline{z_2} - 5z_3| = |3(1-i)^3 + 2(-2+3i) - 5(2i)|$$
$$= |(3-9-4) + i(-9+3+6-10)|$$
$$= |(-10) + i(-10)|$$
$$= \sqrt{(-10)^2 + (-10)^2} = 10\sqrt{2}.$$

Powers of imaginary unit i

Power of imaginary unit $i$are given below:

$$i^0 = 1; i^1 = i; i^{-2} = -1; i^3 = i^2.i = -i;$$
$$i^4 = i^3.i = 1; i^5 = i^4.i = i; i^6 = i^5.i = -1; i^7 = i^6.i = -i$$

One can prove by induction that for any positive integer n

$$i^{4n} = 1; i^{4n+1} = i; i^{4n+2} = -1; i^{4n+3} = -i$$

Hence $i^n \in \{-1,1,-i,i\}$for all integer$n \geq 0$. If $n$is a negative integer, we have

$$i^n = (i^{-1})^{-n} = \left(\frac{1}{i}\right)^{-n} = (-i)^{-n}$$

Example 4.3

$$\boldsymbol{i^{105} + i^{23} + i^{20} - i^{34} = i^{4\times26+1} + i^{4\times5+3} + i^{4\times5} - i^{4\times8+2} = i - i + 1 + 1 = 2}$$

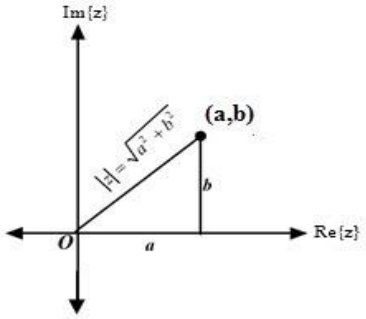


Figure 3: Modulus of a complex number

Polar form of Complex Number and Argument:

It is often useful to exchange Cartesian co-ordinates (x, y) to polar coordinates (r, θ).

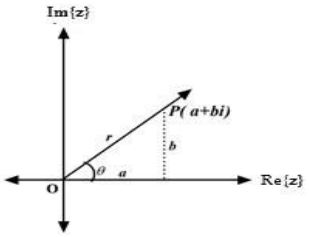


Figure 4: Polar form of complex number

If P is a point in the complex plane corresponding to the complex number (a, b) or$a + ib$, then we see from **Figure 4** that

$$a = r\cos\theta,\ b = r\sin\theta \qquad (1)$$

where, $r = \sqrt{a^2 + b^2}$ is the distance of the point(a, b) from the origin and called the **modulus or absolute value** of$z = a + ib$, denoted by mod z or $|z|$.

On the other hand, if $z \neq 0$, then any number θ *satisfying* the equations (1) is called an **amplitude or argument** of z, and denoted by $arg\ z$. Hence, we can write z in polar form as

$$z = r(\cos\theta + i\sin\theta)$$

Note that, for a given complex number z, $arg\ z$ is not unique. Since adding or subtracting multiples of 2π from θ will result in the arm in Fig. 01 being in the same position, the argument can have many values.

Any choice of length 2π, decided upon in advance, is called the **principal range** and the value of θ is called its principle value.

In our study, we will consider the real number θ the principal argument of z if θ satisfies the equations $0 \leq \theta < 2\pi$ and $-\pi < \theta \leq \pi$. The principal argument of z is usually denoted by Arg z.
Generally, we use the formula for arguments that is

$$\arg z = \theta = \text{Arg}\, z + 2n\pi, \qquad (n = 0, \pm 1, \pm 2, \dots)$$

where $Arg\ z$ may be calculated by the following formula where the quadrant containing the point corresponding to z must be specified,

$$\text{Arg}\, z = \tan^{-1}\left(\frac{b}{a}\right).$$

N.B. Angles measured in an anticlockwise sense are regarded as positive while those measured in a clockwise sense are regarded as negative.
It follows that
$z = a + ib = r(cos\,\theta + i\,sin\,\theta)$, which is called the polar form of the complex number, and $r$and $\theta$are called the polar coordinates.
From the Euler's formula we know, $e^{i\theta} = cos\,\theta + i\,sin\,\theta$.
Hence, the standard polar form of a complex number is $z = re^{i\theta}$.
Some important properties of modulus:

$$|z_1.z_2| = |z_1||z_2|$$
$$|z|^2 = z.\bar{z}$$
$$\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}.$$

Some important properties of argument

$$\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$$
$$\arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2)$$
$$\arg(z^n) = n\arg(z).$$

Example 4.4

Find Im$\{z\}$, where $z = \frac{(1-i)^2}{1+i}$. Hence convert the number from rectangular form to polar form.

Solution:

Given, $z = \frac{(1-i)^2}{1+i} = \frac{(1-i)^2\,(1-i)}{(1+i)\,(1-i)} = \frac{(1-2i-1)(1-i)}{1-i^2}$

$= \frac{-2i+2i^2}{1+1} = -1 - i.$

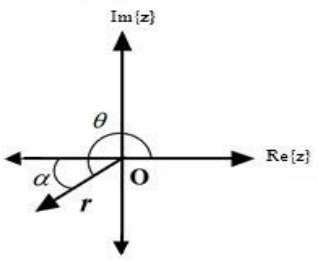
$\therefore$ Im$\{z\} = -1$ and Re$\{z\} = -1$.

Now, $r = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$.

$\alpha = \tan^{-1}\frac{1}{1} = \tan^{-1} 1 = \frac{\pi}{4}$

$\theta = \pi + \alpha = \pi + \frac{\pi}{4} = \frac{5\pi}{4}.$

Therefore, the polar form of $z = \sqrt{2}\, e^{\frac{5\pi}{4}i}$.



Example 4.5

Find the rectangular form of $z = \sqrt{2}e^{i\frac{\pi}{4}}$.

Solution:

Given, $z = \sqrt{2}e^{i\frac{\pi}{4}}$

Here $r = \sqrt{2}$ and $\theta = \frac{\pi}{4}$

We know that $a = r\cos\theta = \sqrt{2}\cos\frac{\pi}{4} = 1$

And $b = r\sin\theta = \sqrt{2}\sin\frac{\pi}{4} = 1$

Hence $z = a + ib = 1 + i$.

Example 4.6

Find the principal argument of $z = \frac{(-1-i)^{12}}{(3+\sqrt{3}\,i)^6\,(-2+2\sqrt{3}\,i)^3}$.

Solution:

Let, $z = \frac{{z_1}^{12}}{{z_2}^6\,{z_3}^3}$, where

$z_1 = -1 - i = \sqrt{2}\ e^{\,i\frac{5\pi}{4}}$ $\quad [r = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$ and $\theta = \pi + \alpha = \pi + \tan^{-1}\frac{1}{1} = \pi + \frac{\pi}{4} = \frac{5\,\pi}{4}\,]$

$z_2 = 3 + \sqrt{3}\,i = 2\sqrt{3}\ e^{\,i\frac{\pi}{6}}$ $\quad [r = \sqrt{12}\ = 2\sqrt{3}$ and $\theta = \tan^{-1}\frac{\sqrt{3}}{3} = \frac{\pi}{6}\,]$

$z_3 = -2 + 2\sqrt{3}\,i = 2^2\ e^{\,i\frac{2\pi}{3}}$ $\quad [r = \sqrt{(-2)^2 + \left(2\sqrt{3}\right)^2}\ = 2^2$ and $\theta = \pi - \alpha = \pi - \frac{\pi}{3} = \frac{2\,\pi}{3}\,]$

Therefore, $z = \frac{\left(\sqrt{2}\ e^{\,i\frac{5\pi}{4}}\right)^{12}}{\left(2\sqrt{3}\ e^{\,i\frac{\pi}{6}}\right)^6\left(2^2\ e^{\,i\frac{2\pi}{3}}\right)^3} = \frac{2^6}{2^6.3^3.2^6}\ \ e^{\,i\,(15\pi-\pi-2\pi)} = \frac{1}{3^3.2^6}\ \ e^{\,i\,12\pi}$.

$\arg z = 12\pi\ = 2\pi + 10\pi$

$\text{Arg}\,z = 2\pi$ or $\text{Arg}\,\text{z} = 0$

Example 4.7

Find the Principal argument of $z_1 = 2e^{-i\frac{\pi}{6}}$ and $z_2 = 15e^{i\frac{15\pi}{2}}$

Solution:

Given, $\arg z_1 = -\frac{\pi}{6}$ or, $-\frac{\pi}{6} + 2\pi = \frac{11\pi}{6}$

$\text{Arg}\,z_1 = \frac{11\pi}{6}$.

$\arg z_2 = \frac{15\pi}{2} = \frac{3\pi}{2} + 6\pi$

$\text{Arg}\,z_2 = \frac{3\pi}{2}$.

De Moivre's Theorem

De Moivre's Theorem is a generalized formula to compute powers of a complex number in its polar form.

Looking at $z = r(\cos\theta + i\sin\theta)$ we can find $(z).(z)$ easily:

$$z^2 = (z).(z) = r^2(\cos2\theta + \text{isin}2\theta)$$

$$z^3 = (z^2).(z) = r^3(\cos3\theta + \text{isin}3\theta)$$

Which brings us to **De Moivre's Theorem**:
If $z = r(\cos\theta + i\sin\theta)$ and n are positive integers then
$z^n = r^n(\cos n\theta + i\sin n\theta) = \left(re^{i\theta}\right)^n = r^n\, e^{in\theta}$
Basically, to find the nth power of a complex number we take the nth power of the absolute value or length and multiply the argument by n.

For finding *m*-th roots of a complex number:

$$z^m = (cos\theta + isin\theta)^m = \cos(m\theta) + i\sin(m\theta)$$

We can also write, $cos\theta + isin\theta = \cos(\theta + 2n\pi) + i\sin(\theta + 2n\pi)$,

$$z^m = (cos\theta + isin\theta)^m = [\cos(\theta + 2n\pi) + i\sin(\theta + 2n\pi)]^m$$
$$= \cos\{m(\theta + 2n\pi)\} + i\sin\{m(\theta + 2n\pi)\}$$

Similarly,

$$z^{\frac{1}{m}} = (\text{cis}\theta)^{\frac{1}{m}} = (cos\theta + i\sin\theta)^{\frac{1}{m}} = [\cos(\theta + 2n\pi) + i\sin(\theta + 2n\pi)]^{\frac{1}{m}}$$
$$= \left[\cos\left(\frac{\theta + 2n\pi}{m}\right) + i\sin\left(\frac{\theta + 2n\pi}{m}\right)\right]$$

where, $n = 0, 1, 2, \ldots, m-1$

Euler's Form:
We can also define this formula according to Euler's formulae as

$$z^m = r^m e^{m\theta i}$$
$$z^{\frac{1}{m}} = r^{\frac{1}{m}} e^{\frac{\theta}{m}i}.$$

Example 4.8

Find all values of z for which $z^3 + 2 - i2\sqrt{3} = 0$ and also locate these values in the complex plane.

Solution: Given, $z^3 + 2 - i2\sqrt{3} = 0$.
Here the numbers of roots are 3.

$z^3 + 2 - i2\sqrt{3} = 0$

or, $z = \left(-2 + i2\sqrt{3}\right)^{\frac{1}{3}}$

or, $z = \left(2^2\ e^{\,i\frac{2\pi}{3}}\right)^{\frac{1}{3}}$

or, $z = \left(2^2\ e^{\,i\left(\frac{2\pi}{3}+2n\pi\right)}\right)^{\frac{1}{3}}$ [As $\theta = \theta + 2n\pi$]

or, $z_n = 2^{\frac{2}{3}}\ e^{\,i\left(\frac{2\pi+6n\pi}{9}\right)}; n = 0,1,2$

when $n = 0$, $z_0 = 2^{\frac{2}{3}}\ e^{\,i\left(\frac{2\pi}{9}\right)}$

when $n = 1$, $z_1 = 2^{\frac{2}{3}}\ e^{\,i\left(\frac{8\pi}{9}\right)}$

when $n = 2$, $z_2 = 2^{\frac{2}{3}}\ e^{\,i\left(\frac{14\pi}{9}\right)}$

The distance of each root from the origin is same as $2^{\frac{2}{3}}$ and the angular distance $\frac{2\pi}{3}$ of two consecutive roots are same.

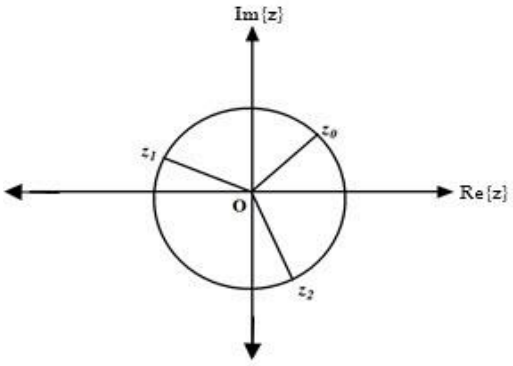


Figure 6

Example 4.9

Describe and graph the locus represented by $1 < |z + i| \leq 2$.

Solution:

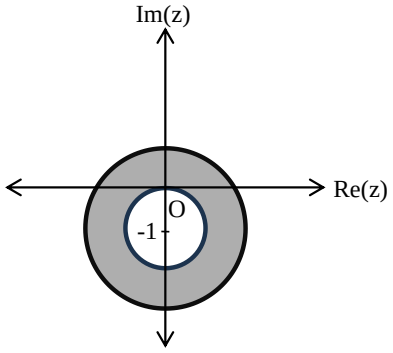
Given, $1 < |z + i| \leq 2$.

or, $1 < |x + iy + i| \leq 2$

or, $1 < |x + i(y + 1)| \leq 2$

or, $1 < \sqrt{x^2 + (y + 1)^2} \leq 2$

or, $1 < (x - 0)^2 + (y - (-1))^2 \leq 2^2$



Exercise Set 4.1

1. Answer the following questions based on the topic discussed

a. Express $\frac{(1+i)^2}{1-i}$ in the form $a + ib$.

b. Evaluate each of the following:

(a) $Re\left\{\frac{1+\sqrt{3}i}{1-i}\right\}$, (b) $\left|\frac{z}{\bar{z}}\right|$, (c) $\text{Im}\left\{\frac{z}{\bar{z}}\right\}$.

c. Convert the following numbers into polar form:

i. $z = -1 + i$,

ii. $z = -3 - \sqrt{3}i$,

iii. $z = \frac{(1-i)^2}{1+i}$,

iv. $z = \left(\frac{1-i}{1+i}\right)^{18}$.

d. Convert the following numbers into rectangular form:

$z = \sqrt{3}e^{i\frac{\pi}{3}}$ and $z = 2e^{i\frac{\pi}{4}}$.

e. Find the principal argument for the following:

i. $z = (-1 - i)^4$,

ii. $z = \left(-2 + 2\sqrt{3}\, i\right)^3$,

iii. $z = \frac{(1+i)^3}{(1-i)}$,

f. Find all values of z for the following equations and locate these values in the complex plane:

i. $z^2 + 9 = 0$,

ii. $z^3 - \sqrt{3} - i = 0$.

iii. $z^3 = -i$.

iv. $z^4 - 1 = 0$.

g. Describe and graph the locus represented by each of the following:

i. $|z + 2i| > 4$.

ii. $1 < |z - 2 + i| \leq 3$.

iii. $\text{Im}\{z^2\} = 9$.

iv. $|z - 1| \leq 1$.

v. $\text{Re}\{z^2\} < 4$.

vi. $\frac{\pi}{6} \leq \arg z \leq \frac{\pi}{3}$.

2. **Problems from Book(**Complex Analysis with Applications – Dennis G. Zill)

a. If $z_1 = 2 + 4i$, $z_2 = -3 + 8i$, $z_3 = 2 - 3i$ and $z_4 = -4 - 6i$, then find

i. Locate z_1, z_2, z_3 and z_4 in Argand Diagram
ii. $z_1 z_2$
iii. $z_1 + z_2$
iv. $|z_1 \bar{z}_2 - z_3|$
v. $\frac{z_3}{z_4} + \frac{z_1}{z_2}$
vi. $\frac{1}{(z_2)^3} + z_1$

b. Write the given number in the form $a + ib$.

i. $(5 - 9i) + (2 - 4i)$
ii. $i(4 - i) + 4i(1 + 2i)$
iii. $3(4 - i) - 3(5 + 2i)$
iv. $i(5 + 7i)$
v. $\frac{i}{1 + i}$
vi. $\frac{(3 - i)(2 + 3i)}{1 + i}$
vii. $3i + \frac{1}{2 - i}$
viii. $\frac{(4 + 5i) + 2i^3}{(2 + i)^2}$
ix. $\frac{10 - 5i}{6 + 2i}$
x. $\frac{2 - 4i}{3 + 5i}$
xi. $\frac{(5 - 4i) - (3 + 7i)}{(4 + 2i) + (2 - 3i)}$
xii. $\left(\frac{1}{2} - \frac{1}{4}i\right)\left(\frac{2}{3} + \frac{5}{3}i\right)$
xiii. $\frac{2 - 3i}{(4 + i)}$
xiv. $\frac{(1 + i)(1 - 2i)}{(2 + i)(4 - 3i)}$

c. Let $z = -6 - 5i$. Find

i. $\mathrm{Re}\left(\frac{1}{z}\right)$
ii. $\mathrm{Re}(iz)$
iii. $\mathrm{Re}(z^2)$
iv. $|z|$
v. $\mathrm{Im}(2z + 4\bar{z} - 4i)$
vi. $(1 + \mathrm{i})\mathrm{z}$
vii. $\mathrm{Im}(\bar{z}^2 + z^2)$
viii. $\mathrm{Im}(iz)$.

d. Find the modulus of the following Complex number:

i. $z = (1 - i)^2$
ii. $z = \frac{2i}{3 - 4i}$
iii. $z = i(2 - i) - 4\left(1 + \frac{1}{4}i\right)$
iv. $z = \frac{1 - 2i}{1 + i} + \frac{2 - i}{1 - i}$

e. Find the principal argument for the following Complex numbers:

i. $z = 1 + i$
ii. $z = \frac{1 - i}{(1 + i)^2}$
iii. $z = -1 + \sqrt{3i}$
iv. $z = -5 + 5i$

f. Convert the following Complex numbers into polar form:

i. $z = 2$
ii. $z = -\sqrt{3} + i$
iii. $z = -1$
iv. $z = 1 + i$
v. $z = -3i$
vi. $z = -2 - 2\sqrt{3}i$
vii. $z = 1 + i$
viii. $z = \dfrac{1+i}{(1-i)^2}$

g. For the following equations, find all values of z and locate these values in the complex plane:

i. $\mathrm{z} = (-1)^{\frac{1}{4}}$
ii. $\mathrm{z} = (-1+i)^{\frac{1}{5}}$
iii. $\mathrm{z} = (1+i)^{\frac{1}{5}}$
iv. $z = \left(-1+\sqrt{3}i\right)^{\frac{1}{2}}$
v. $\mathrm{z} = (8)^{\frac{1}{3}}$
vi. $z = \left(-1-\sqrt{3}i\right)^{\frac{1}{4}}$

h. Describe and graph the locus represented by each of the following:

i. $|z+3i| > 4,$
ii. $1 < |z-5+i| \leq 2,$
iii. $|z-1+2i| > 5,$
iv. $|z| \leq 4.$

Mappings

Geometrical Representation:

To draw a curve of a complex variable (x, y) we take two axes, i.e., one real axis and the other imaginary axis. A number of points (x, y) are plotted on z-plane, by taking different values of z (different value of *x* and *y*). The curve C is drawn by joining the plotted points. The diagram obtained is called the **Argand diagram.**

Transformation:

For every point (x, y) in the z-plane, the relation $w = f(z)$ defines a corresponding point (u, v) in the w-plane. We call this "transformation or mapping of z-plane into w-plane". If a point z_0 maps into the point w_0, $w_0$is known as the image of z_0.

If the point $P(x, y)$ moves along a curve *C* in z-plane, the point $P'(u, v)$ will move along a corresponding curve C_1 in the w-plane. We, then, say that a curve C in the z-plane is mapped into the corresponding curve C_1 in the w- plane by the relation $w = f(z)$.

Translation, Rotation, and reflection are the standard transformations. Terms such as **translation, rotation,** and **reflection** are used to convey dominant geometric characteristics of certain mappings.

Translation

$w = z + C,$

where,

$$C = a + ib$$
$$z = x + iy$$

$$w = u + iv$$

Hence, $u + iv = x + iy + a + ib$

So, $u = x + a$ and $v = y + b$

$$x = u - a \text{ and } y = v - b$$

On substituting the values of x and y in the equation of the curve to be transformed, we get the equation of the image in the w-pane.

As an example, the mapping $w = z + 1$ where $z = x + iy$, can be thought of as a translation of each point of z one unit to the right.

Example 4.10

Let the rectangular region R in the z-plane, which is bounded by the lines

$$x = 0, y = 0, x = 2, y = 1.$$

Determine the region R' of the w-plane into which R is mapped under the transformation $w = z + 1$.

Solution:

Given $w = z + 1$

or $u + iv = (x + 1) + iy.$

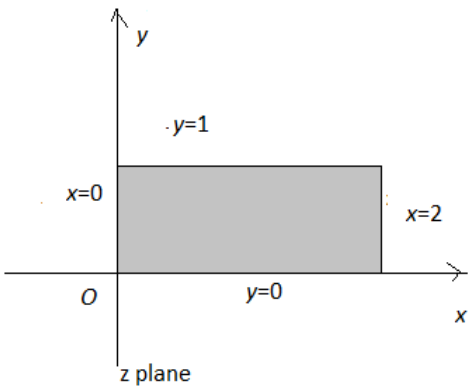
Hence $u = x + 1$ and $v = y.$

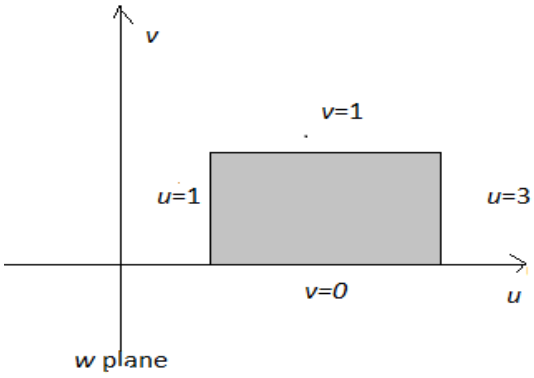
when $x = 0,\ u = 1,$

$y = 0,\ v = 0$

$x = 2,\ u = 3,$

$y = 1,\ v = 1$





u

Rotation:

The mapping $w = iz$ where $z = re^{i\theta}$ and $i = e^{i\frac{\pi}{2}}$, can be thought of as a rotation of the radius vector for each non-zero point z through a right angle about the origin in the counterclockwise direction.

Example 4.11

Let the rectangular region R in *the* z-plane, which is bounded by the lines

$$x = 0, y = 0, x = 2, y = 1.$$

Determine the region R' of the w-plane into which R is mapped under the transformation

$$w = iz.$$

Solution:

Given $w = iz$

or, $u + iv = -y + ix.$

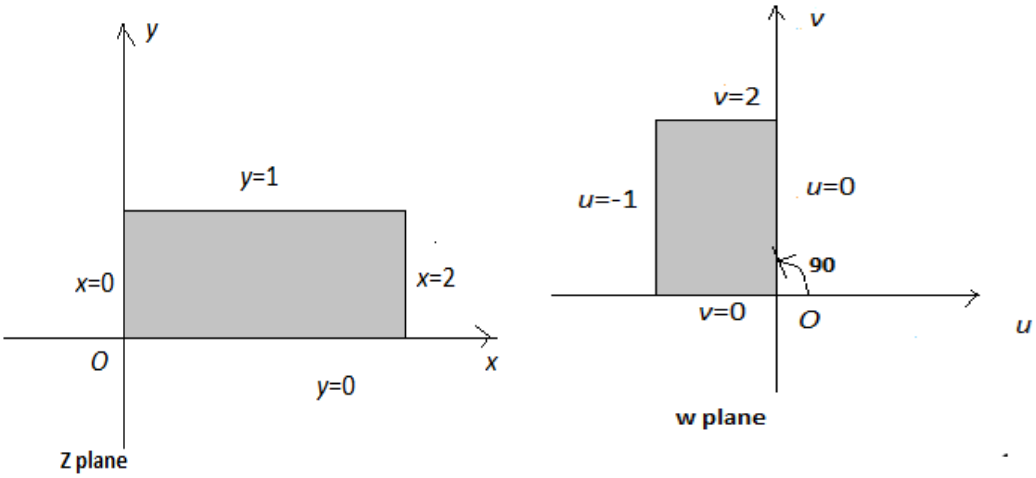
Hence $u = -y$ and $v = x.$

when $x = 0,\ v = 0,$

$y = 0,\ u = 0$

$x = 2,\ v = 2,$

$y = 1,\ u = -1$



Reflection:
The mapping $w = \bar{z}$ transforms each point of $z = x + iy$ into its reflection in the real axis.

Example 4.12
Let the rectangular region R in *the* z-plane, which is bounded by the lines

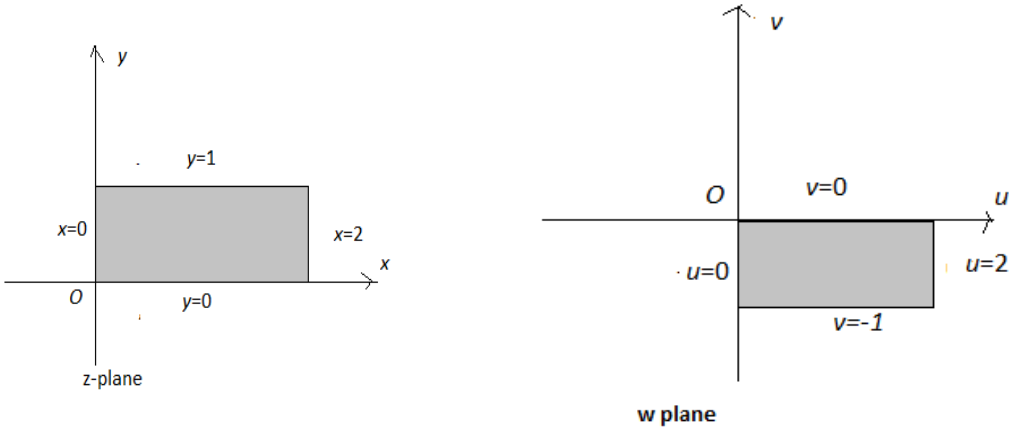
$$x = 0, y = 0, x = 2, y = 1.$$

Determine the region R' of the *w*-plane into which R is mapped under the transformation

$$w = \bar{z}\,.$$

Solution:
Given $w = \bar{z}$
or, $u + iv = x - iy$.
Hence $u = x$ and $v = -y$.

when $x = 0,\ u = 0,$
$y = 0,\ v = 0$
$x = 2,\ u = 2,$
$y = 1,\ v = -1$



Example 4.13
Given triangle T in the z-plane with vertices at $-1 + 2i, 1 - 2i$ and $1 + 2i$. Determine the triangle T' of the *w*-plane into which T is mapped under the transformation $w = \sqrt{2}e^{\frac{\pi i}{4}}z$.

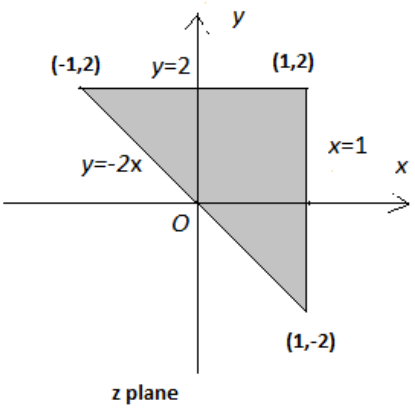
Solution:

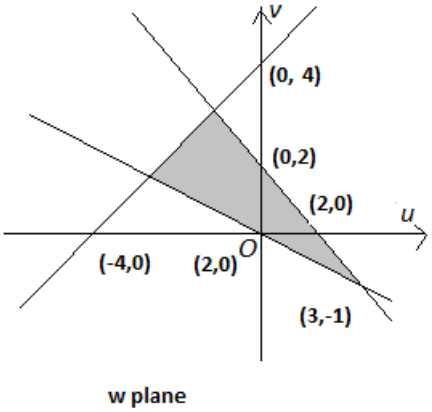
Given $w = \sqrt{2}e^{\frac{\pi i}{4}}z = (1+i)(x+iy)$
or, $u + iv = (x-y) + i(x+y)$.
Hence $u = x - y$ and $v = x + y$.

When
$x = 1$,
$y = 2$
$y = -2x$

The vertices of the triangle are
$-1+2i, 1-2i, 1+2i$. Hence the sides are
$x = 1$, $y = 2$ and $y = -2x$.

$u = 1 - y, v = 1 + y \quad => u + v = 2$
$u = x - 2, v = x + 2 \quad => u - v = -4$
$u = 3x, v = -x \quad => u + 3v = 0$





Exercise Set 4.2

1. Answer the following questions based on the topics studied above:

a. Let the rectangular region R in *the* z-plane, which is bounded by the lines $x = 2, y = 0, x = 5$ and $y = 4$. Determine the region R' of the w-plane into which R is mapped under the following transformations:

i. $w = 2z - (2 + 3i)$

ii. $w = \frac{1}{2}e^{\frac{\pi i}{2}}z + 2i$

iii. $w = \sqrt{2}e^{\frac{\pi i}{4}}z - (1 - i)$

iv. $w = e^{i\pi}z + 3 + i$

v. $w = \frac{1}{\sqrt{2}}e^{\frac{\pi i}{4}}z + 1 - 3i$

b. Given triangle T in the z -plane with vertices at $1, 1 - 3i$ and $3 - i$. Determine the triangle T' of the w-plane into which T is mapped under the following transformations:

(i) $w = 3z + 1 - 3i$, (ii) $w = iz + 3 + 2i$,

(iii) $w = (1 + 2i)z - i$, (iv) $w = \frac{1}{2}e^{\frac{\pi i}{2}}z - 4$.

c. A triangle in the z-plane is defined by the straight lines
$y = 3x$, $y = -x$ and $x = -1$.
Mapping of the triangle from z- plane to w-plane is defined by, $w = \frac{1}{\sqrt{2}}e^{-\frac{i\pi}{4}}z - i$.
Find the triangle in w-plane by answering the following question:
i. Find $u(x, y)$ and $v(x, y)$ so that, $w = u(x, y) + i\, v(x, y)$.

ii. Use the relations between x and y in the equation of the sides of the triangle in the z-plane to find the relationship between u and v in the w-plane.
iii. Draw the new triangle in w-plane and identify the vertices.

2. Problems from book (A first course in complex analysis with applications, Dennis G. Zill)

a. Find the image S' of the square S with vertices at $1 + i, 2 + i, 2 + 2i$, and $1 + 2i$ under the linear mapping $T(z) = z + 2 - i$
b. Find the image of the rectangle with vertices $-1 + i, 1 + i, 1 + 2i$, and $-1 + 2i$ under the linear mapping $f(z) = 4i \cdot z + 2 + 3i$
c. Find the image of the triangle with vertices 0, 1, and i under the given linear mapping $w = f(z)$:

i. $f(z) = z + 3i$
ii. $f(z) = z + 2 - i$
iii. $f(z) = 3iz$
iv. $f(z) = 1 + iz$
v. $f(z) = 2z - i$
vi. $f(z) = 4 - 3iz + 1 - 3i$
vii. $f(z) = \frac{1}{\sqrt{2}} e^{\frac{i\pi}{4}} z + \frac{1}{2} - \frac{i}{2}$
viii. $f(z) = \sqrt{2} e^{i\frac{\pi}{4}} z + 2i$
ix. $f(z) = -2z + 1 - \sqrt{3}i$
x. $f(z) = (3 - 2i)z + 1$